RUTURAJ DESAI

Philadelphia, PA | +1 267-439-6176 | rutura jd@engineering.upenn.edu | linkedin.com/in/ruturajsdesai/ | Portfolio

Education

University of Pennsylvania **Expected: May 2027**
Master's in Mechanical Engineering - Mechatronic & Robotic Systems *Philadelphia PA, USA*
- Relevant Coursework: Mechatronic System Design, Robotics, Feedback Control Design & Analysis, MEMS/NEMS

University of Mumbai **May 2023**
Bachelor's in Mechanical Engineering *Mumbai, India*
- Relevant Coursework: Vehicle Dynamics, CAE, CAD/CAM, Machine Design, Material Sciences, Fluid Mechanics

Technical Skills

Core Competencies: Multibody Dynamics, Modelling & Simulation, CAD, FEA, DFMEA, PFMEA, CFD, GD&T
Design & Simulation: Simpack, MSC Adams, MATLAB, Inspire, ANSYS Structural, SolidWorks, AutoCAD, CATIA
Process Workflow & Automation: GIT, JavaScript, Python, C/C++, Microsoft Excel
Fabrication & Manufacturing: 3D Printing, Laser Cutting, Water Jet, Fixture Routing, Lathe, Milling, CMM

Experience

MID-WEST FORGE CORPORATION **May 2026 – Aug 2026**
Mechanical Design, Rebuild & Manufacturing Intern *Cleveland OH, USA*
- Integrated laser detection systems, replacing traditional mechanical switches in 2 axle processing lines
- Optimized cycle times for load & stack FANUC, ABB robots, improving codes for control modules integrated with PLC
- Assisted in the rebuild of heavy machinery & setup of different axle designs from Ford & GM in the forging process

DASSAULT SYSTÈMES **Jul 2023 – Jul 2025**
Industry Process Consultant - Multibody Dynamics | World-Wide Tech Sales *Pune, India*
- Facilitated Simpack deployment as a part of WW Tech Sales team in the USA, India, Italy, Germany, China & S.Korea
- Delivered Simpack database solutions to automotive, truck & motorsport clients via PoCs enabling new OEM workflows
- Drove a successful PoC & deployment for an Italian motorsport team, correlating 55+ metrics for F1/F3 models

Vehicle Dynamics & Powertrain Apprentice | Database Development
- Refined 100+ components, automated KnC/full-vehicle loadcases, version-controlled with GIT for automotive databases
- Handled the development of 4 high-fidelity, 2-stage reduction eDrive gearbox models in single & dual e-motor setups
- Built 2 customized KnC loadcase sets, automated through node.js, cutting simulation wall clock times by 45%

TEAM REDSHIFT RACING INDIA-BAJA SAE **Jan 2020 – May 2023**
Team Captain *Mumbai, India*
- Led a team of 45 interdisciplinary undergraduates, ensuring a successful season with a top 2 finish amongst 196 teams
- Cut prototype cost by 55% through lean manufacturing, JIT inventory, make-buy decisions & architecture redesign
- Secured funding from 8 major industry partners, by pitching to management, boosting financial capacity & future R&D

Vehicle Dynamics Head & System Integration
- Overhauled suspension design, reducing unsprung mass by 21% to enable transition from propeller to chain-driven 4WD
- Redesigned the steering system, by incorporating an integrated rack & achieved a targeted turning radius of 1.9 meters
- Maximized weight reduction to 11% in uprights & wheel hubs through topology optimization using Altair Inspire

UNIVERSITY OF MUMBAI **Aug 2022 – Dec 2022**
Research Assistant | K.J. Somaiya College of Engineering *Mumbai, India*
- Modeled functionally graded beams using Euler–Bernoulli theory with linear, quadratic, & exponential modulus
- Validated analytical models with FEA simulations, achieving less than 3% deviation under point & distributed loads
- Achieved a 20% tip deflection drop and smoother displacement using functionally graded material design

Projects & Activities

- Design & Fabrication process of a Micro-ElectroMechanical System (MEMS) Gyroscope (2026)
- Motion planning & optimization for a 7 DoF Franka-Emika Robot using forward & inverse kinematics (2025)
- ESP32 based autonomous battle bot with PID-controlled differential drive & autonomous/Wi-Fi control (2025)
- Design, Analysis and Fabrication of a Suspension & Steering System of a 4WD All-Terrain Vehicle (2023)
- Self-Driving Cars Specialisation - University of Toronto (Online)

Publications

- Stress Analysis of Functionally Gradient Beam & Finite Element Analysis using ANSYS Workbench. *IJIRSET* https://www.ijirset.com/doi:10.15680/IJIRSET.2022.1112060 (2022)